

DP-800 LAB 02



Connect to AdventureWorksLT

```
1  -- Verify key tables in AdventureWorksLT
2  SELECT TOP (5) ProductID, Name, ListPrice
3  FROM SalesLT.Product;
4
5  SELECT TOP (5) ProductCategoryID, Name
6  FROM SalesLT.ProductCategory;
```

83 % No se encontraron problemas.

Resultados Mensajes

	ProductID	Name	ListPrice
1	680	HL Road Frame - Black, 58	1431,50
2	706	HL Road Frame - Red, 58	1431,50
3	707	Sport-100 Helmet, Red	34,99
4	708	Sport-100 Helmet, Black	34,99
5	709	Mountain Bike Socks, M	9,50

	ProductCategoryID	Name
1	4	Accessories
2	22	Bib-Shorts
3	30	Bike Racks
4	31	Bike Stands
5	1	Bikes

Build JSON output from product data

```
1  SELECT
2      ProductID,
3      Name,
4      Color,
5      ListPrice
6  FROM SalesLT.Product
7  WHERE Color IS NOT NULL
8  ORDER BY ListPrice DESC
9  FOR JSON PATH;
```

100 % No se encontraron problemas.

Resultados Mensajes

	JSON_F52E2B61-18A1-11d1-B105-00805F49916B
1	[{"ProductID":749,"Name":"Road-150 Red, 62","Col...

Create nested JSON with product categories

1	SELECT
2	p.ProductID,
3	p.Name AS ProductName,
4	p.ListPrice,
5	JSON_OBJECT(
6	'CategoryID': pc.ProductCategoryID,
7	'CategoryName': pc.Name
8	) AS Category
9	FROM SalesLT.Product AS p
10	INNER JOIN SalesLT.ProductCategory AS pc
11	ON p.ProductCategoryID = pc.ProductCategoryID
12	ORDER BY p.ListPrice DESC
13	FOR JSON PATH;

100 % No se encontraron problemas.

Resultados Mensajes

JSON_F52E2B61-18A1-11d1-B105-00805F49916B	
1	[{"ProductID":749,"ProductName":"Road-150 Red, 6...

Combine JSON with a CTE and window function

1	WITH RankedProducts AS (
2	SELECT
3	p.ProductID,
4	p.Name AS ProductName,
5	pc.Name AS CategoryName,
6	p.ListPrice,
7	ROW_NUMBER() OVER (
8	PARTITION BY pc.ProductCategoryID
9	ORDER BY p.ListPrice DESC
10	) AS PriceRank
11	FROM SalesLT.Product AS p
12	INNER JOIN SalesLT.ProductCategory AS pc
13	ON p.ProductCategoryID = pc.ProductCategoryID
14	WHERE p.ListPrice > 0
15	)
16	SELECT
17	ProductID,
18	ProductName,
19	CategoryName,
20	ListPrice,
21	PriceRank
22	FROM RankedProducts
23	WHERE PriceRank <= 3
24	ORDER BY CategoryName, PriceRank;

% No se encontraron problemas.

Resultados Mensajes

ProductID	ProductName	CategoryName	ListPrice	PriceRank
855	Men's Bib-Shorts, S	Bib-Shorts	89.99	1
856	Men's Bib-Shorts, M	Bib-Shorts	89.99	2
857	Men's Bib-Shorts, L	Bib-Shorts	89.99	3
876	Hitch Rack - 4-Bike	Bike Racks	120.00	1
879	All-Purpose Bike Stand	Bike Stands	159.00	1
871	Mountain Bottle Cage	Bottles and Cages	9.99	1
872	Road Bottle Cage	Bottles and Cages	8.99	2
870	Water Bottle - 30 oz.	Bottles and Cages	4.99	3
996	HL Bottom Bracket	Bottom Brackets	121.49	1
995	ML Bottom Bracket	Bottom Brackets	101.24	2
994	LL Bottom Bracket	Bottom Brackets	53.99	3
948	Front Brakes	Brakes	106.50	1
907	Rear Brakes	Brakes	106.50	2
712	AWC Logo Cap	Caps	8.99	1
952	Chain	Chains	20.24	1
877	Bike Wash - Dissolver	Cleaners	7.95	1

Output the ranked products as JSON

1	WITH RankedProducts AS (
2	SELECT
3	p.ProductID,
4	p.Name AS ProductName,
5	pc.Name AS CategoryName,
6	p.ListPrice,
7	ROW_NUMBER() OVER (
8	PARTITION BY pc.ProductCategoryID
9	ORDER BY p.ListPrice DESC
10	) AS PriceRank
11	FROM SalesLT.Product AS p
12	INNER JOIN SalesLT.ProductCategory AS pc
13	ON p.ProductCategoryID = pc.ProductCategoryID
14	WHERE p.ListPrice > 0
15	)
16	SELECT
17	ProductID,
18	ProductName,
19	CategoryName,
20	ListPrice,
21	PriceRank
22	FROM RankedProducts
23	WHERE PriceRank <= 3
24	ORDER BY CategoryName, PriceRank
25	FOR JSON PATH, ROOT('TopProducts');

100 %
No se encontraron problemas.

Resultados

Mensajes

JSON_F52E2B61-18A1-11d1-B105-00805F49916B	
1	["TopProducts":{"ProductID":855,"ProductName":...}]

Parse a JSON array into rows

1	DECLARE @ProductUpdates NVARCHAR(MAX) = N'[
2	{"ProductID": 680, "NewPrice": 1250.00},
3	{"ProductID": 706, "NewPrice": 1450.00},
4	{"ProductID": 707, "NewPrice": 38.99}
5	];
6	
7	SELECT
8	ProductID,
9	NewPrice
10	FROM OPENJSON(@ProductUpdates)
11	WITH (
12	ProductID INT '\$.ProductID',
13	NewPrice DECIMAL(10,2) '\$.NewPrice'
14	);

100 %
No se encontraron problemas.

Resultados

Mensajes

	ProductID	NewPrice
1	680	1250.00
2	706	1450.00
3	707	38.99

Join parsed JSON with existing data

1	DECLARE @ProductUpdates NVARCHAR(MAX) = N'[
2	{"ProductID": 680, "NewPrice": 1250.00},
3	{"ProductID": 706, "NewPrice": 1450.00},
4	{"ProductID": 707, "NewPrice": 38.99}
5	];
6	
7	SELECT
8	p.ProductID,
9	p.Name,
10	p.ListPrice AS CurrentPrice,
11	updates.NewPrice,
12	updates.NewPrice - p.ListPrice AS PriceDifference
13	FROM SalesLT.Product AS p
14	INNER JOIN OPENJSON(@ProductUpdates)
15	WITH (
16	ProductID INT '\$.ProductID',
17	NewPrice DECIMAL(10,2) '\$.NewPrice'
18	) AS updates
19	ON p.ProductID = updates.ProductID;

100 %
No se encontraron problemas.

Resultados

Mensajes

	ProductID	Name	CurrentPrice	NewPrice	PriceDifference
1	680	HL Road Frame - Black, 58	1431.50	1250.00	-181.5000
2	706	HL Road Frame - Red, 58	1431.50	1450.00	18.5000
3	707	Sport-100 Helmet, Red	34.99	38.99	4.0000

Enlace al repositorio de GitHub

[Repositorio GitHub](#)